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중국 음악교육에서 인공 지능 관련 음악 교육 연구: 2017~2025년 문헌을 바탕으로 하는 체계적 분석

Research on Artificial Intelligence-Related Studies in Chinese Music Education: A Systematic Analysis Based on the Literature From 2017 to 2025

장개림*
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Abstract The rapid development of artificial intelligence has attracted increasing attention within the Chinese music education community. This study aims to provide a systematic review of research in this field. Using bibliometric methods, it conducts a multi-dimensional analysis of publication trends, research topics, research objects, institutional distribution, journal affiliation, and research methods. The findings indicate that the field has evolved through stages of initial exploration, rapid expansion, and deeper reflection. Research mainly focuses on comprehensive discussions of artificial intelligence and its educational applications, with higher education institutions serving as the primary contributors. In terms of methodology, theoretical research occupies a dominant position, while empirical studies remain relatively limited. Overall, research on artificial intelligence in Chinese music education is still largely at the stage of theoretical development and requires stronger interdisciplinary collaboration and more empirical evidence to achieve deeper integration and sustainable advancement.

Key words: artificial intelligence, music education, bibliometrics, China, educational technology, interdisciplinary research

초록 인공지능 기술의 급속한 발전은 중국 음악교육 학계의 높은 관심을 불러일으켰다. 본 연구는 해당 분야의 연구 동향과 구조적 특징을 체계적으로 고찰하는 것을 목적으로 한다. 이를 위해 서지계량학적 방법을 활용하여 발표 추이, 연구 주제, 연구 대상, 기관 분포, 학술지 소속, 연구 방법 등 다차원적 분석을 수행하였다. 분석 결과, 본 분야는 초기 탐색 단계에서 급속한 확장 단계를 거쳐 최근에는 심화 성찰 단계로 발전해 왔음을 확인하였다. 연구 주제는 인공지능의 이론적 논의와 교육적 적용을 중심으로 전개되었으며, 연구 수행 주체는 주로 고등교육기관에 집중되어 있다. 연구 방법 측면에서는 이론 중심 연구가 우세한 반면, 실증적 연구는 상대적으로 부족한 것으로 나타났다. 전반적으로 중국 음악교육 분야의 인공지능 연구는 아직 이론적 구축 단계에 머물러 있으며, 향후 학제 간 협력과 실증 연구의 확대가 요구된다.

주제어: 인공지능, 음악교육, 계량서지학적 분석, 중국, 교육공학, 학제 간 융합

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Ⅰ. Introduction

Artificial intelligence, recognized as one of the most transformative technological developments of the 21st century, has fundamentally reshaped educational paradigms across global contexts. In the field of music education, the penetration and integration of artificial intelligence technology presents an unprecedented development trend. From the innovative breakthrough of intelligent composition system to the widespread application of personalized learning platform, from the technical realization of virtual music tutor to the precise optimization of adaptive assessment tools, AI technology is fundamentally redefining the theoretical boundaries and practical paths of music teaching and learning. This technology-driven educational reform not only opens up unprecedented possibilities for the personalization, intelligence and popularization of music education, but also provides strong technical support for the innovative development of music talent training mode.

The international academic community has built a relatively mature and diversified research system for the application of artificial intelligence in music education, covering multiple core aspects such as technology development, teaching application, effect evaluation, and ethical thinking. In recent years, the revolutionary breakthrough of generative artificial intelligence technology has brought new opportunities and challenges to the field of music education. A large-scale systematic review found that the artificial intelligence music generation technology based on deep learning has reached a level of complexity and sophistication comparable to or even partially surpassing the level of human creation, providing extremely rich creative tools and teaching resources for music education practice (Civit et al., 2022). This technological breakthrough not only significantly expands the boundaries of the possibilities of music creation, but also opens up a new development path for the innovative practice of music education. Advanced machine learning analysis methods were used to examine usage patterns of global online music learning platforms during the COVID-19 pandemic (Knapp et al., 2023). The results clearly revealed the key role played by artificial intelligence technology in promoting the digital transformation of music education, especially the excellent adaptability and strong flexibility shown in responding to sudden public health crises. The remarkable effectiveness of artificial intelligence in music teaching plan formulation and teaching quality assessment was further verified through an empirical study. By accurately comparing the quality differences between teaching plans generated by AI and plans formulated by professional music teachers, the study found that the AI system even demonstrated superior performance to human teachers in certain specific dimensions, providing solid empirical support for the in-depth practical application of AI-assisted music education (Cooper, 2024).

In the context of Chinese academia, the rise and vigorous development of AI music education research is closely related to the national strategic policy orientation. In 2017, the Chinese government officially promulgated and implemented the ‘New Generation Artificial Intelligence Development Plan’, a milestone national strategic document, which clearly proposed to systematically promote the innovative application and deep integration of artificial intelligence technology in the field of education. This major policy measure not only provides strong policy support and sufficient development opportunities for music education AI research, but also points out a clear development direction for related academic research. The implementation of this policy not only provides abundant financial support, but also establishes a clear research orientation and value pursuit for the entire academic community. Driven by this policy, related research has sprung up in large numbers, covering multiple levels from basic theoretical exploration to cutting-edge practical applications, which fully reflects the Chinese academic community's high attention and active participation in this emerging cross-disciplinary field (Hu et al., 2024).

Based on the above-mentioned practical needs and academic gaps, this study uses rigorous bibliometric research methods to conduct a comprehensive, systematic, multi-dimensional and in-depth analysis of relevant high-quality literature officially published by the Chinese academic community from 2017 to the first quarter of 2025. The study aims to comprehensively and deeply reveal the historical development trajectory, core characteristics and potential challenges of China's artificial intelligence music education research by building a scientific and complete analytical framework. Through this systematic analysis from a macro perspective, it can not only provide the academic community with a complete panoramic view of the development of this field, but also provide a solid scientific basis and empirical support for the scientific determination of future research directions, the precise implementation of policy formulation, and the effective promotion of practical applications.

Ⅱ. Theoretical Background

1. The development trajectory of the application of artificial intelligence technology in music education

The integration of artificial intelligence technologies within music education contexts has demonstrated progressive evolution, characterized by increasing sophistication and diversification of applications (Linnenluecke et al., 2020). Its evolution process profoundly reflects the high degree of integration and mutual promotion between advanced technology and educational needs. From the perspective of technical foundation, it mainly covers core technical fields such as machine learning, deep learning, natural language processing, computer vision, and speech recognition. The continuous maturity and optimization of these cutting-edge technologies have laid a very solid technical foundation and innovation platform for the comprehensive intelligent transformation of music education.

Recent technological advances have enabled diverse applications across multiple domains. Deep neural network-based systems have achieved notable progress in automatic music generation and personalized learning environments. These developments have facilitated adaptive instructional approaches, as evidenced by studies demonstrating AI-driven tutoring systems capable of responding to individual learner needs (Della Ventura, 2022; Huang et al., 2019).The system can not only scientifically and accurately evaluate the quality and creative level of learners' works, but also provide highly personalized improvement suggestions and optimization plans, successfully realizing a closed-loop feedback mechanism from simple evaluation to comprehensive guidance.

2. Analysis of international research status and development trends

International scholarship demonstrates interdisciplinary characteristics with emphasis on empirical methodologies and practical applications. Recent studies have examined AI implementation across diverse educational contexts, from K-12 settings to professional music training (Amankwah-Amoah et al., 2024; Casal-Otero et al., 2023).

3. The uniqueness, development advantages and challenges of Chinese studies

China has unique cultural background advantages and development characteristics in the field of artificial intelligence music education research. This uniqueness not only constitutes an important development advantage, but also brings special challenges and constraints. From a positive perspective, China has extremely rich and profound traditional music and cultural resources, which provide unique research objects and practical scenarios for the innovative application of AI technology. The complexity, diversity and uniqueness of traditional Chinese music provide an extremely rich and high-quality data source for the training and optimization of AI algorithms, and also provide a unique cultural resource foundation for the innovative practice of music education. On the other hand, China has the world's largest and most complete education system and the largest number of learners, which provides an extremely broad experimental platform and application scenarios for large-scale AI technology application and effect verification.

An in-depth study was conducted on the innovative model and implementation strategy of Chinese university music education based on AI technology, providing an important reference framework and practical guidance for the digital transformation and intelligent upgrading of China's higher music education (Wei et al., 2022). This study fully combines the actual situation and special needs of China's higher education, proposes localization strategies and implementation paths for the application of AI technology in music education, and particularly emphasizes the balance and coordination between the application of advanced technology and the inheritance of excellent culture. The development status, main characteristics and existing problems of Chinese music education research from 2007 to 2019 were comprehensively analyzed through a systematic literature review, providing an important historical foundation and reference basis for a deeper understanding of the traditions and developmental trends of Chinese music education research (Yang & Welch, 2023). This study systematically reveals the unique characteristics and obvious deficiencies of Chinese music education research, and provides important background information and decision-making support for the scientific application and localized development of AI technology in Chinese music education.

However, China also faces many severe challenges and constraints in this research field. The first is the significant disconnect between theoretical research and practical application. Most current research still remains at the surface level of concept introduction and technical analysis, and there is a serious lack of in-depth empirical research and application effect verification. The emergence of this phenomenon is closely related to the traditional preferences and evaluation orientation of Chinese academic research, as well as the limitations of actual application conditions and technical thresholds. The second is the serious lack of interdisciplinary cooperation. The synergy between related disciplines such as music education, computer science, cognitive psychology, and learning science has not yet been fully exerted, and disciplinary barriers still exist. This lack of interdisciplinary cooperation not only limits the depth and breadth of research, but also seriously affects the practical application value and social influence of research results. Finally, there is the problem of relatively low internationalization. Exchanges and cooperation with international advanced research are still limited, which to a certain extent affects the cutting-edge, innovative and international influence of research.

4. China's application of AI technology in music education research

Existing Chinese literature explores AI applications from multiple perspectives. Research encompasses primary and secondary education contexts, emphasizing technology-enhanced pedagogical strategies, as well as higher education and professional training focusing on specialized instrumental instruction. Additionally, scholars have examined platform development and philosophical implications of AI integration (He & Ren, 2024; Wei et al., 2022).

Ⅲ. Research Methods

1. Study design

This investigation employs a bibliometric-systematic literature review (B-SLR) methodology, integrating systematic literature review protocols with quantitative bibliometric analyses to provide comprehensive mapping of the research landscape of high-quality literature related to artificial intelligence music education published by Chinese academia from 2017 to the first quarter of 2025. The research design strictly follows the relevant requirements and academic norms of the PRISMA 2020 guidelines to ensure the scientificity, systematicity and repeatability of the literature screening and analysis process, and provide a solid guarantee for the reliability and validity of the research results.

2. Literature search strategy

This study uses CNKI as the primary data source to ensure systematic and comprehensive literature collection.The main retrieval database is China National Knowledge Infrastructure (CNKI), which is the most comprehensive and authoritative academic database in China. The retrieval time span is set from January 2017 to March 2025, and the retrieval language is strictly limited to Chinese.

The search term combination strategy uses a combination of subject terms and keywords:

Keywords: ‘artificial intelligence’ or ‘intelligent education’ or ‘machine learning’ or ‘deep learning’ or ‘AI’ or ‘intelligentization’

Conjunction: ‘music education’ or ‘music teaching’ or ‘music learning’ OR ‘music course’ or ‘music training’

3. Literature screening criteria

1) Inclusion criteria

Literature meeting the following inclusion criteria was selected for analysis. The temporal scope encompassed publications from January 1, 2017 to March 31, 2025, focusing exclusively on Chinese-language scholarly works. Included materials comprised academic journal articles, doctoral and master's dissertations, and conference proceedings that explicitly addressed artificial intelligence applications in music education contexts. Both theoretical and empirical investigations were incorporated, provided they demonstrated adherence to fundamental academic standards and contributed substantive scholarly value to the field.

2) Exclusion criteria

Studies were systematically excluded based on specific criteria designed to ensure methodological rigor and relevance. Duplicate publications or highly similar works were removed to prevent redundancy. Literature that mentioned artificial intelligence only superficially in abstracts without substantive discussion was excluded, as were purely technical papers lacking clear educational application contexts. Additionally, studies with unavailable full texts, those exhibiting poor scholarly quality, or those displaying evidence of academic misconduct were eliminated from the sample.

4. Data extraction and coding

This study constructed a scientific and comprehensive multi-dimensional data extraction framework, comprising six core dimensions, informed by prior bibliometric analyses in AI music education (Kim et al., 2024), with adaptations to suit the Chinese research context:

- Time dimension: publication year, publication quarter, development stage
- Thematic dimension: research topic classification, core keywords, and innovation point identification
- Object dimension: research object type, educational stage positioning, target group characteristics
- Institutional dimension: first author institution type, geographical distribution, and collaborative network
- Journal dimensions: journal type, subject affiliation, impact factor level
- Methodological dimensions: type of research method, data collection method

The data extraction process followed a systematic coding procedure with multiple rounds of review and validation to ensure the accuracy, consistency, and reliability of data extraction.

5. Data analysis methods

This study uses a variety of data analysis methods, using professional software such as SPSS 28.0, Excel 2019 and Python for data processing, statistical analysis and visualization. Analytical procedures included descriptive statistical analysis encompassing frequency distributions, proportional calculations, and measures of central tendency. Temporal patterns were examined through time series analysis, growth rate calculations, and identification of developmental phases. Cross-dimensional analyses explored correlations between variables, conducted association tests, and performed comparative assessments. Content analytical methods facilitated thematic synthesis, and in-depth interpretation of research characteristics (Linnenluecke et al., 2020).

Ⅳ. Research Results

To provide an overview of the analytical framework and the distribution of the reviewed literature, Table 1 summarizes the main analytical dimensions and key findings of this study, including publication phases, research topics, educational stages, institutional types, and research methods.

<Table 1> Overview of research literature on AI in Chinese music education (2017-2025 Q1, N=83)

Analytical dimension	Main categories	Number of studies (n)	Percentage (%)
Publication phase	Initial exploration (2017-2019)	9	10.8
	Rapid development (2020-2022)	26	31.3
	Deepen reflection (2023-2025Q1)	48	57.8
Research topics	Comprehensive exploration and development of artificial intelligence	29	34.9
	Applied research across educational stages	23	27.7
	Application of instrumental and vocal music teaching	11	13.2
	AI-driven model and system innovation	9	10.8
	Ethical issues and challenges	9	10.8
	Teacher training and teaching innovation	1	1.2
	Generative AI and its new applications	1	1.2
Educational stages	Higher education	16	19.2
	Secondary education	7	8.4
	Primary education	5	6.0
	Early childhood and special education	3	3.6
	Unspecified education stage	52	62.7
Institution type	Higher education institutions	62	74.7
	Secondary education institutions	14	16.9
	Primary education institutions	5	6.0
	Educational research institutions	2	2.4
Research methods	Theoretical research	78	94.0
	Quantitative research	3	3.6
	Qualitative research	2	2.4

1. Analysis of the number of papers and publication trends

Initial exploratory stage (2017-2019): This stage shows the starting characteristics and infrastructure nature of this research field. A total of only 9 papers have been published, accounting for 10.8% of the total. Although the number is relatively small, year-on-year changes exhibited substantial fluctuations, indicating an unstable yet dynamic development pattern in the early stage. The research content of this stage mainly focuses on the definition of basic concepts, the exploration of technical possibilities and the preliminary construction of theoretical frameworks, reflecting the academic community's initial cognitive process and tentative attention to this emerging interdisciplinary field. From the perspective of research quality, early studies are mostly review and conceptual studies with relatively limited theoretical depth, but they have laid an important cognitive foundation and theoretical preparation for subsequent in-depth research.

Rapid development stage (2020-2022): This stage marks the beginning of a period of substantial development and large-scale expansion in this research field. A total of 26 high-quality papers have been published, accounting for 31.3% of the total. Among them, the number of publications in 2020 has significantly jumped to 12, with a year-on-year growth rate of 200%, becoming an important turning point in the development of this field. This stage of development has obvious external driving characteristics, and the global digital transformation of education caused by the COVID-19 pandemic has become an important catalyst. The large-scale popularization of online education models has not only prompted scholars to pay more attention to the application potential of artificial intelligence technology in music education, but also created real needs and application scenarios for the practical application of related technologies. From the perspective of the evolution of research content, more specialized research on specific application scenarios began to appear in this stage, such as intelligent piano teaching systems, online music education platform design, and music learning effect evaluation, which fully reflects the important shift in the focus of research from theoretical exploration to practical application.

Deepening reflection stage (2023-2025): This stage presents the characteristics of maturity and quality development trend of this research field. 48 high-quality papers have been published, accounting for 57.8% of the total. Among them, the number of publications in 2024 reached a historical peak of 27, with a year-on-year growth rate of 285.7%. This phenomenon is obviously deeply influenced by the revolutionary breakthroughs in generative AI technology in 2022-2023. The widespread application of generative AI tools such as ChatGPT, DALL-E, and Midjourney has re-stimulated the academic community's enthusiasm and deep thinking on the application of artificial intelligence in education. Significantly different from the previous two stages, the research in this stage shows stronger critical thinking and reflective characteristics, and begins to pay more attention to deep-seated issues such as ethical issues, cultural bias, educational equity, and technology dependence in the application of AI technology, which fully reflects the significant expansion of research vision and substantial improvement in thinking depth.

<Table 2> Statistics of literature publications and analysis of their phased characteristics from 2017 to 2025

Years	Number of publications	Percentage (%)	Cumulative quantity	Cumulative percentage (%)	Year-on-year growth rate (%)	Development stage	Key features
2017	1	1.2	1	1.2	-	Initial exploration	Concept introduction
2018	4	4.8	5	6.0	300.0	Initial exploration	Theoretical foundation
2019	4	4.8	9	10.8	0.0	Initial exploration	Cognitive construction
2020	12	14.5	21	25.3	200.0	Rapid development	Application exploration
2021	7	8.4	28	33.7	-41.7	Rapid development	Rational adjustment
2022	7	8.4	35	42.2	0.0	Rapid development	Steady growth
2023	7	8.4	42	50.6	0.0	Deepen reflection	Quality improvement
2024	27	32.5	69	83.1	285.7	Deepen reflection	Explosive growth
2025Q1	14	16.9	83	100.0	-	Deepen reflection	Continued popularity
Total	83	100.0					

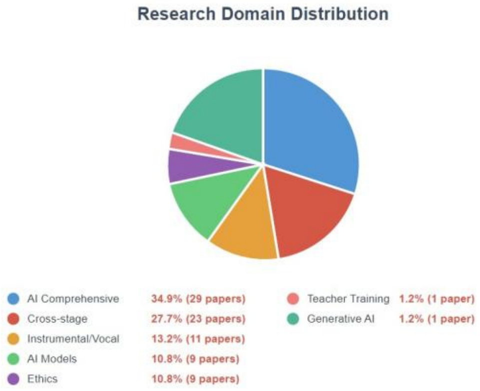
These temporal patterns demonstrate distinct developmental characteristics across three phases. The initial period showed limited research activity, the middle phase marked substantial growth in publication volume, while the recent period exhibits both quantitative expansion and qualitative maturation, as indicated by increased representation in core academic journals.The observed publication trends reflect broader contextual influences.From the macro analysis of the overall development trend, China's AI music education research shows obvious policy sensitivity and technology follow-up characteristics. Every major policy or key technological breakthrough will trigger a corresponding research boom, which not only reflects the keen insight and rapid adaptability of the Chinese academic community, but also exposes the passive responsiveness and potential trend-following tendency of the research to a certain extent. Publications in the first quarter of 2025 (14 articles), according to the current trend, the annual publication volume is expected to continue to surpass the historical record in 2024, which clearly shows the continued popularity and strong development potential of this research field.

2. Research topic classification and in-depth content analysis

Table 3 and Figure 1 illustrate the distribution of research topics across seven major categories. Comprehensive exploration and development of artificial intelligence emerged as the predominant focus with 29 articles (34.9%), followed by applied research across educational stages with 23 articles (27.7%). Application of instrumental and vocal music teaching accounted for 11 articles (13.2%), while AI-driven model and system innovation and ethical issues and challenges represented 9 articles each (10.8%). Smaller proportions were observed in teacher training and teaching innovation alongside generative AI applications (1 article each, 1.2%).

<Table 3> Research topic classification statistics

Research topics	Number of documents	Proportion (%)	2017-2019	2020-2022	2023-2025
Comprehensive exploration and development of artificial intelligence	29	34.9	5	10	14
Applied research across educational stages	23	27.7	3	7	13
Application of instrumental and vocal music teaching	11	13.2	1	5	5
AI-driven model and system innovation	9	10.8	0	1	8
Ethical issues and challenges	9	10.8	0	2	7
Teacher training and teaching innovation	1	1.2	0	1	0
Generative AI and its new applications	1	1.2	0	0	1
Total	83	100.0	9	26	48



[Figure 1] Research topic distribution

These distributional patterns reveal notable imbalances in research priorities and concerning is that there are very few studies on two extremely important topics, teacher training and teaching innovation, and generative AI and its new applications, with only one article each. This phenomenon deserves in-depth analysis and high attention. The serious lack of research on teacher training profoundly reflects that current research focuses too much on technology itself and seriously neglects the key factor of people. This is an extremely serious structural defect and development shortcoming. In the promotion and application of any educational technology, teachers are always the most critical and decisive factor. The lack of systematic attention to teacher training will seriously affect the actual application effect and long-term development prospects of AI technology in music education. The serious lack of research on generative AI profoundly reflects the obvious lag of Chinese academia in tracking the development of international cutting-edge technologies. Considering the research enthusiasm of generative AI in the international academic community and its huge application potential in music creation and teaching, this lack of research needs to be improved and strengthened.

3. In-depth analysis of research subjects and education stages

Table 4 presents the distribution of research across educational stages. The data reveal that 52 articles (62.7%) did not specify a particular educational stage. Among studies with clearly defined educational contexts, higher education research constituted the largest proportion with 16 articles (19.2%), followed by secondary education with 7 articles (8.4%), primary education with 5 articles (6.0%), and early childhood and special education with 3 articles (3.6%). The ambiguity in the reference of research objects is the most serious and far-reaching structural problem in this research field. More than 60% of the studies (52 articles, 62.7%) failed to clearly define the specific educational stage. This serious ambiguity has multiple complex manifestations and deep-seated reasons. From the specific manifestations, it mainly includes the following aspects: First, the widespread use of extreme vague terms such as ‘primary and secondary schools’, ‘schools of all levels and types’, and ‘different educational stages’ without clear distinction and positioning of specific stages; second, there is a general lack of clear positioning of application objects in theoretical research, and only general discussions are conducted from an abstract technical perspective; third, the so-called cross-stage research seriously lacks hierarchical analysis, and does not fully consider and deeply study the special needs, condition differences and development laws of different educational stages.

<Table 4> Distribution of educational stages in AI music education studies

Educational stage	Number of documents	Percentage(%)	Main research content	Problems
Higher education	16	19.2	Professional skills training, AI literacy education	Lack of interdisciplinary integration
Secondary education	7	8.4	Music appreciation, basic skills training	Relatively limited scope of application
Primary education	5	6.0	Music enlightenment and interest cultivation	Theory and practice are out of touch
Early childhood and special education	3	3.6	Early music development, special needs support	Lack of empirical validation
Unspecified education stage	52	62.7	Discussion on generalization theory	Serious lack of targeting
Total	83	100.0		

From the perspective of a systematic analysis of the deep-level causes, this serious problem is caused by the combined effects of multiple complex factors. First, there are significant limitations in the disciplinary background and practical experience of researchers. Many researchers seriously lack rich practical experience and in-depth professional understanding of music education, and their understanding of the unique characteristics, development needs and implementation conditions of different educational stages is obviously insufficient, which directly leads to the lack of necessary pertinence and professionalism in research. Secondly, there is a tendency for theoretical research to be overly abstract. Researchers often excessively pursue the so-called universality and generality of theories, and seriously ignore the importance and particularity of specific application scenarios. Thirdly, there are obvious deficiencies in the journal review system and quality control mechanism. Some journals obviously do not have sufficient requirements for the rigor and standardization of research design, and lack effective quality control mechanisms and academic standards. Finally, there are problems with the orientation of research incentive mechanisms and evaluation systems. The current academic evaluation system tends to focus on the so-called theoretical and innovative nature of research, while the requirements for practicality, pertinence and application value are relatively low or even ignored.

In the research on clear education stages, there are relatively more higher education research and the overall quality is relatively high, which is closely related to the natural advantages and resource endowments of higher education institutions in academic research. AI music education research at the higher education stage comprehensively covers many important aspects such as listening training, sight-singing and ear training, vocal performance, instrumental teaching, music creation, music theory, etc. In recent years, it has also actively expanded to emerging important topics such as the cultivation of AI technical literacy of music students and the construction of interdisciplinary innovation capabilities. However, even in this relatively mature and developed research field, there is still an obvious problem of insufficient depth of interdisciplinary integration, and the organic combination of AI technology application and music education theory is not deep and systematic enough.

Research at the secondary education stage focuses more on actual classroom applications and teaching practices, and has a strong practice orientation and application value. This type of research mainly focuses on the specific application of AI technology in music appreciation teaching, basic music theory teaching, instrument performance teaching, music creation guidance, etc., but the scope of application is relatively limited and lacks systematicity and comprehensiveness. In-depth analysis found that research at the secondary education stage focuses more on the simple utilization of AI technology tools, but pays less attention to how AI technology can be deeply integrated with music education concepts, teaching methods and evaluation systems.

Primary school education research mainly focuses on the development of AI-assisted music teaching strategies and the training of basic music skills. This type of research is innovative and forward-looking, but there is a general problem of a serious disconnect between theory and practice. Many studies have proposed seemingly idealized application models and implementation plans, but they lack sufficient consideration of actual application conditions and scientific verification of application effects, and their practicality and operability need to be significantly improved.

Although the number of studies on early childhood and special education is the smallest, it has extremely high development potential and important social value. This type of research mainly appears in 2024, which fully reflects the continuous expansion and deepening of the research field. The application of AI in early childhood education mainly focuses on the function of music enlightenment education and aesthetic education, while the research in special education focuses on the support and promotion effect of AI technology on music learning for students with special needs. Although this type of research has important social value and broad application prospects, it is still seriously lacking in empirical verification and effect evaluation, and most research still remains at the level of theoretical discussion and conceptual analysis.

4. In-depth analysis of research institution distribution and academic ecology

As the absolute main force in this research field, higher education institutions have contributed nearly three-quarters of the research results. This distribution pattern fully reflects the natural advantages and core position of higher education institutions in academic research, and also deeply reflects the basic characteristics of the highly academic nature of this research field. However, there are significant differences in academic performance among different types of higher education institutions, which deeply reflects the structural characteristics of China's higher education system and the complexity of this research field.

<Table 5> Comprehensive analysis of the distribution of research institution types

Institution type	Number of documents	Percentage(%)	Key benefits	Core deficiency
Higher education institutions	62	74.7	Rich resources and concentrated talents	Out of touch with practice
Secondary education institutions	14	16.9	Strong practical application orientation	Limited research conditions and capabilities
Primary education institutions	5	6.0	Rich experience in basic education	Serious lack of research capacity
Educational research institutions	2	2.4	Obvious advantages in policy research	Seriously insufficient quantity
Total	83	100.0		

<**Table 6**> Distribution of higher education institution types in AI-related music education research (N=62)

Institution type	Number of documents	Percentage(%)	Key benefits	Core deficiency
Comprehensive university	21	33.9	Interdisciplinary resource advantages	Lack of professional depth
Normal university	20	32.3	Rich experience in educational practice	Weak technical foundation
Music conservatory	14	22.6	Strong professional authority	Limited technological innovation capabilities
Art university	7	11.3	Artistic innovation with unique perspective	Insufficient theoretical construction ability
Total	62	100.0		

Comprehensive universities have demonstrated unique and significant comprehensive advantages in this research field. They not only occupy a leading position in the number of publications, but also have relatively outstanding research quality and innovation capabilities, and a relatively high degree of international cooperation. This significant advantage mainly stems from the extremely rich interdisciplinary resource endowment of comprehensive universities, which can effectively integrate the high-quality strengths of multiple disciplines such as computer science, education, musicology, psychology, and cognitive science, and provide comprehensive and systematic disciplinary support and resource guarantees for AI music education research. However, the research of comprehensive universities also has the obvious problem of relatively insufficient professional depth. Due to the lack of deep professional background and practical accumulation in music education, their research often focuses on technical exploration, and the attention paid to the special laws, deep-seated problems and core essence of music education is obviously insufficient.

Normal universities have become an important force and key player in this research field by virtue of their profound accumulation and rich experience in the field of education. The significant advantages of normal universities are mainly reflected in their deep understanding of educational laws and rich accumulation of educational practice experience. Their research focuses more on core educational issues such as the organic integration of AI technology and educational concepts, the innovative development of teaching methods, and the reconstruction and changes of teacher-student relationships. However, normal universities are relatively weak in terms of technological foundation and innovation capabilities, which limits their academic contributions in technological innovation and system development. From the perspective of international cooperation, the degree of internationalization of normal universities is obviously low, which may be closely related to their relatively closed academic environment and limited international exchange resources.

The academic performance of music conservatories presents complex and contradictory characteristics. On the one hand, as a professional and authoritative institution for music education, music conservatories have irreplaceable professional authority and disciplinary advantages in this research field, and their research is often highly professional, targeted and of practical guidance value; on the other hand, the number of research conducted by music conservatories is relatively small, and their performance in technological innovation and interdisciplinary cooperation is relatively limited. The emergence of this phenomenon is closely related to the traditional disciplinary structure, academic culture and development concept of music conservatories. Traditionally, music conservatories have paid more attention to the cultivation of artistic practice and performance skills, and have paid relatively little attention to emerging technologies and interdisciplinary research, lacking the institutional mechanisms and cultural foundation for effective cooperation with science and engineering fields.

Art colleges have the lowest level of participation, a phenomenon that deserves in-depth reflection and systematic analysis. Although art colleges have unique advantages and irreplaceable value in artistic innovation and creative thinking, they are relatively weak in theoretical construction, academic research, and interdisciplinary cooperation. This situation not only limits the academic influence and voice of art colleges in this research field, but also makes this research field lack in-depth participation and innovative contributions from an artistic perspective.

Although the participation of basic education institutions is limited in number, it has important practical value and application significance. Although the research of secondary education institutions is relatively low in academic influence, it has a strong practice orientation and application value, and can provide real application scenarios, practical experience and effect feedback. Primary education institutions participate even less, but their research is often highly targeted and operational. The main reasons for the low participation of basic education institutions include multiple factors such as limited research conditions, insufficient research capabilities, lack of incentive mechanisms, and imperfect evaluation systems.

The serious lack of participation of educational research institutions is an important defect and development shortcoming of the academic ecology in this research field. Educational research institutions should play an important leading role in educational policy research, theoretical construction, international comparison, development planning, etc., but their current participation is obviously insufficient, which seriously affects the policy orientation, strategic foresight and international influence of this research field.

5. Analysis of journal distribution and interdisciplinary development

In-depth analysis of journal distribution not only comprehensively reflects the disciplinary characteristics and publication platform distribution of this research field, but more importantly, it profoundly reveals the degree of disciplinary integration, development trends and future directions. This has extremely important theoretical significance and practical value for a deep understanding of the academic characteristics, development laws and future directions of this research field.

The distribution pattern of journals deeply reflects the disciplinary characteristics and development trends of this field. Literary, historical and philosophical journals dominate, publishing more than half of the research results. This phenomenon has dual significance. On the positive side, it reflects the active attention and deep involvement of the music discipline in the application of AI technology, ensuring the professionalism and authority of the research. On the negative side, it reflects that research in this field is overly dependent on the traditional disciplinary framework, and the degree of true interdisciplinary integration is still limited.

Among the literature, history and philosophy journals, music and dance journals are the absolute main force, which is in line with the professional characteristics of this field. The research published in this type of journal is often highly professional and targeted, but it is also easy to form a closed discipline, which limits the expansion of research horizons. The proportion of core journals in music and dance journals is relatively high, indicating that important research results in this field are mainly published through professional journals.

The importance of educational and social science journals is becoming increasingly prominent, accounting for nearly one-fifth of the total number of publications, and the proportion of core journals is the highest. The research of this type of journal focuses more on core issues such as the application effect of AI technology in music education, innovation of teaching models, and construction of learning theories, and has a strong educational orientation.

Although the number of technical science journals is small, they are of great significance. The emergence of such journals marks the beginning of the field breaking through the boundaries of traditional disciplines and moving towards true interdisciplinary research. More importantly, technical science journals have the highest proportion of core journals and the strongest international influence, indicating that high-quality interdisciplinary research has begun to be highly recognized by the academic community.

6. In-depth analysis of research methods

The absolute dominance of theoretical research (94.0%) is the most prominent feature of this field and also the most serious structural problem. The emergence of this phenomenon has complex reasons and far-reaching impacts. From a positive perspective, theoretical research has laid a conceptual foundation and cognitive framework for the development of this field. In emerging interdisciplinary fields, it is necessary to take theory first. However, the negative impact of over-reliance on theoretical research is more obvious: research lacks empirical support and it is difficult to verify the validity of the theory; research is divorced from practical needs and has limited practical value; research is highly homogenized and lacks innovation.

<Table 7> Types of research methods used in AI music education studies

Research methods	Number of documents	Percentage(%)
Theoretical research	78	94.0
Quantitative research	3	3.6
Qualitative research	2	2.4
Total	83	100.0

<Table 8> Types of theoretical research in AI music education studies (N=78)

Research methods	Number of documents	Percentage(%)
Concept analysis	39	50.0
Theory construction	31	39.7
Philosophical thinking	8	10.3
Total	78	100.0

In the internal structure of theoretical research, conceptual analysis research accounts for the largest proportion, which mainly focuses on the explanation of AI-related technologies and general discussion of application conditions in music education. In-depth analysis found that this type of research has serious duplication problems, many research contents are highly similar, and lack original contributions.

Although the quality of theoretical construction research is relatively high, it also has obvious defects. This type of research is mainly committed to building application models, implementation paths, and evaluation frameworks for AI in music education, and is innovative and forward-looking. However, the vast majority of theoretical construction research lacks an effective verification mechanism, and the proposed theoretical models often lack practical verification.

Although there are fewer philosophical speculation studies, they are of a high academic level. This type of research mainly focuses on deep-seated issues such as the ethical risks, value conflicts, and educational philosophy impact of AI technology applications, and has high theoretical value and ideological depth.

The serious lack of empirical research is the most prominent structural problem in this field. There are only 3 quantitative studies and 2 qualitative studies, accounting for less than 6% in total. This situation is in sharp contrast to advanced international research and has seriously restricted the academic reputation and actual influence of this field.

Ⅴ. Discussion

China's AI music education research presents a distinct “policy-driven” development feature. The promulgation of the “New Generation Artificial Intelligence Development Plan” in 2017 directly gave rise to the rise of research in this field. The digital transformation of education driven by the COVID-19 pandemic in 2020 and the breakthrough of generative AI technology in 2024 triggered two research climaxes respectively. This development model can quickly mobilize academic resources and form research hotspots, but it also brings about a short-term tendency, utilitarian orientation and bandwagoning in research.

From the perspective of the evolution of research content, Chinese scholars have shown a clear tendency to “put theory first”. The proportion of theoretical research, 94.0%, is not only far higher than the international average, but also exceeds the general rules of emerging interdisciplinary disciplines. The formation of this tendency is influenced by both cultural traditions and practical conditions. However, excessive theory-first also brings serious problems: the disconnection between theory and practice affects the practical value of research results, and theories that lack practical verification often become empty.

Ⅵ. Conclusion

This study reveals the development trajectory of artificial intelligence music education in China from the introduction of the concept to theoretical construction and reflective deepening through a systematic bibliometric analysis of research from 2017 to 2025. The study finds that Chinese scholars have initially established a theoretical framework with local characteristics in the field of artificial intelligence music education, but still face structural challenges such as vague research object positioning, insufficient interdisciplinary integration, and a lack of empirical research.From the perspective of academic development, the current stage is at a critical juncture of transition from theoretical exploration to practical application. Future research should achieve breakthroughs at three levels: at the methodological level, it is necessary to construct a mixed research paradigm combining quantitative analysis and qualitative research to verify the effectiveness of theoretical constructs with empirical data; at the disciplinary integration level, it is necessary to break down traditional disciplinary barriers and establish a mechanism for in-depth collaboration between musicology, computer science, cognitive psychology, and educational technology; At the international level, it is necessary to actively participate in global academic dialogue while maintaining cultural subjectivity, and promote the transformation of Chinese experience into international academic discourse. The application of artificial intelligence technology in music education is not only a technical issue, but also a value issue concerning the inheritance and innovation of music culture.Faced with the opportunities and challenges brought by generative artificial intelligence, music educators need to strike a balance between technological rationality and humanistic spirit, fully leveraging the advantages of artificial intelligence in personalized learning and intelligent assessment while adhering to the fundamental mission of music education to cultivate well-rounded personalities and preserve cultural spirit. Only in this way can artificial intelligence technology truly become a powerful support for the high-quality development of music education, rather than merely a tool for technological demonstration.

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